Bradley Alexander Cougar Stiles

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PORTFOLIO

OBJECTIVE

Engineering Technology M.S. specializing in **energy/trajectory co-optimized control systems**, with four years of professional experience spanning industry engineering and collegiate teaching, graduate research deploying controllers on physical hardware, and leadership of a 100-member engineering organization. Seeking a controls, robotics, automation, or manufacturing engineering role in Texas. Available January 2027.

EDUCATION

Texas A&M University — College Station, TX
M.S., Engineering Technology (MSET), Thesis Option — Controls, AI, Project Management | GPA 3.5/4.0 — *Dec 2026*
B.S., Multidisciplinary Engineering Technology (MXET) — Mechatronics; Minor, Embedded Systems Integration — *Dec 2024*

PUBLICATIONS

A. E. Duron, B. A. C. Stiles, and D. A. Torck, "RIPTIDE: Robot Inspecting Parts to Increase Development Efficiency," *Proc. 22nd Int. Conf. Ubiquitous Robots (UR)*, College Station, TX, 2025, pp. 333–339.
B. A. C. Stiles, "Predictive Co-Optimization of Trajectory and Hybrid Power Split for Electric Ground Delivery Vehicle," M.S. thesis, Texas A&M University, 2026 (pending).

RESEARCH & ENGINEERING PROJECTS

MARCO: Energy-Optimal Autonomous Ground EV — Texas A&M University — College Station, TX
Graduate Research Assistant — Controls & Autonomy (M.S. Thesis) — *Jan 2025 – Dec 2026*

- Designed, tuned, and deployed a **model predictive controller (MPC)** that **co-optimizes path planning and power split in a single controller**, resolving energy-optimal routing, obstacle avoidance, and energy management together rather than as separate issues.
- Implemented control over a **hybrid battery / supercapacitor energy storage system**, routing transient torque and current spikes to the supercapacitor so the battery sees smooth sustained draw.
- Validated in simulation and closed the loop on **physical hardware** at **15–20 Hz** against a perception-bound update rate.
- Trained and deployed a **custom computer-vision neural network** for onboard perception, running GPU-accelerated **PyTorch** inference on an **NVIDIA Jetson Orin** across **four ZED 2i stereo camera pairs** at VGA resolution, 15 Hz.
- Architected the **ROS 2** sensor network integrating multi-camera CV and mapping data, centimeter-level GPS and localization, and kinematic and power-draw telemetry from the drive, steering, and hybrid power systems.
- Directed undergraduate research teams across wiring, chassis structure, software architecture, and controls testing.
- Collaborated with a PhD candidate developing an **offline reinforcement learning** controller on the same vehicle, coordinating shared hardware and test conditions to support a direct comparison of model-based and learned controls.

RIPTIDE: Automated Pick-and-Place / 3D-Scanning QC Robot — Additive at Scale — College Station, TX
Mechatronics Engineer — Senior Capstone Team — *Nov 2023 – Dec 2024*

- Built a mobile automated 3D-scanning solution around a **UR5 robotic arm** with a dual end effector handling both part manipulation and metrology.
- Designed a custom claw and rotary axis integrated with a **Creaform Black Elite 3D scanner** to capture **quality-control** and reverse-engineering data on mass-produced parts.
- Reduced the manual QC cycle from **50–60 minutes** to **6–10 minutes** depending on setup, a **5–10x throughput gain**.
- Fabricated custom components in-house using **additive manufacturing**, including 3D printing and injection molding.

INDUSTRY EXPERIENCE

Tracks North America — Industrial Hydraulic & Electric Vehicles, Equipment, and Controllers — College Station, TX
Mechatronics Engineer — Intern, then Co-op (part-time, concurrent with senior capstone) — *Nov 2023 – Dec 2024*

- Developed, with the engineering team, a **1,000 lb, 4 ft tracked electric construction vehicle**.
- Individually designed and built a **wireless, single-handed, long-range controller** for remote operation of the electric vehicle.
- Enhanced and updated industrial-grade radio-controlled **hydraulic construction equipment**.

Byrd Innovations — Automated Board Repair Robots College Station, TX
Project Manager, Lead Engineer *Nov 2022 – Aug 2023*

- Led a team of engineers to design and manufacture a robot that desolders, cleans, and resolders any combination of the encoder wheel and buttons on a computer mouse in **about 5 minutes**.
- Supported a senior engineering team on a companion repair robot that detects and replaces surface-mounted chips and heatsinks on digital mining boards.

Professional Teaching & Leadership Experience

Texas A&M University — ETID Department, ESET Program College Station, TX
Graduate Assistant — Teaching, ESET 333 Product Development *Jan 2025 – Dec 2026*

- Taught engineering project management and product development to **300+ undergraduate engineers** across four semesters and two summer terms, covering multiple sections each term.
- Delivered curriculum spanning industry research, context and functional block diagrams, waterfall project management, reverse engineering, time value of money and cash flow analysis, go-to-market strategy, and technical presentation skills.
- Developing course rubrics and training incoming graduate teaching assistants ahead of graduation.

Aggie Robotics — ETID Department, Texas A&M University College Station, TX
Vice President(2022–2023), President (2024–2025); Graduate Advisor (2025–2026) *Aug 2021 – Dec 2026*

- Grew and directed a ∼**100-member** volunteer engineering organization fielding multiple **VEX U** competition robots in parallel, from design through international competition.
- Raised and managed a ∼**$50K annual operating budget** through corporate and departmental sponsorship; owned allocation across parts, tooling, travel, and event hosting.
- Led the organization to four **VEX U World Championship** awards in two years: **Design** and **A3 Safety** (2026), **Workforce Readiness** and **Energy** (2025).
- Coordinated parallel build schedules against academic calendars to sustain retention and delivery on an entirely unpaid team; transitioned the organization from waterfall planning to **Agile** sprints tracked in Notion.
- Organized **1,000+ hours** of community volunteer service over five years; hosted regional HS/MS VEX competitions as event lead and represented the organization through press and televised interviews, sponsor reporting, and recruiting media.

Technical Skills

Controls & Autonomy: Model predictive control (MPC), PID, MIMO nonlinear control, feedback linearization, Lyapunov stability analysis, dynamic system simulation, path and motion planning, graph search (A*, Dijkstra), obstacle avoidance, state estimation and localization, centimeter-level GPS, sensor fusion, hardware-in-the-loop testing

Energy & Power Systems: Hybrid battery / supercapacitor energy storage, power-split optimization, transient current and torque buffering, energy management strategy, power-draw telemetry and instrumentation

Perception, AI & Machine Learning: PyTorch, CNN training and dataset labeling, object detection, edge GPU inference, camera calibration, stereo depth and point clouds, SLAM and occupancy grid mapping

Software: C, C++, Python, MATLAB / Simulink, ARM 64-bit assembly, Verilog HDL, ROS 2, Linux, Git, HTML/CSS, object-oriented programming (OOP)

Embedded & Electrical: ARM architecture, Raspberry Pi, Arduino, ESP32, TI development boards, PLC, digital logic, Multisim, power and motor control, serial protocols (UART, I2C, SPI), PWM, ADC/DAC, bench test and measurement (oscilloscope, logic analyzer, multimeter, function generator), soldering

Robotics Hardware: UR5 / PolyScope programming, forward and inverse kinematics, end-effector and gripper design, servo / stepper / BLDC drives, encoders, 3D scanning metrology (Creaform / VX), hydraulic and pneumatic actuation

CAD & Manufacturing: SolidWorks, Fusion 360, Creo, FEA, 3D printing (Bambu / Prusa / Cura), injection molding, CNC router, G-code and CAM toolpaths, machining

Project Management: Waterfall and Agile, Scrum and Kanban, sprint planning, backlog management, stand-ups, Gantt charts, work breakdown structure (WBS), responsibility assignment matrix (RAM), network logic diagrams, IDEF functional modeling, budget ownership, risk management, sponsor and stakeholder communication, Notion

Coursework

Controls, AI & Analysis (graduate): ENTC 689 Applied Control and Optimization · ENTC 689 Generative AI: Introduction and Applications · ENTC 641 Data Analysis, Simulation and Experimental Methods · STAT 651 Statistics in Research I

Project Management (graduate): ENTC 629 Leadership and People Management in Technology Enterprises · ISEN 665 Management of Engineering Systems · ISEN 663 Engineering Management Control Systems

Undergraduate: ENTC 489 Robotic Systems Design · ESET 462 Control Systems · MXET 300 / 400 Mechatronics I & II · MXET 375 Applied Dynamic Systems · ESET 369 Embedded Systems Software · ESET 349 Microcontroller Architecture